

Area Between Curves



Area between a curve and the x-axis

- 1 Find the area of the region bounded by $y = 4 - x^2$ and the x-axis.
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Area between two curves (vertical strips)

- 2 Find the area between $y = x^2$ and $y = x + 2$.
- 3 Find the area between $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.
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Regions requiring horizontal strips

- 4 Find the area of the region bounded by $x = y^2$ and $x = y + 2$, integrating with respect to y .
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Regions with multiple intersection points

- 5 Find the total area between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$, accounting for the curves crossing at $x = \frac{\pi}{4}$.
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Area using the more efficient variable of integration

- 6 Find the area enclosed by $y^2 = 4 - x$ and $x = 0$, choosing the more convenient variable of integration.
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Setting up but not evaluating (AP-style)

- 7 Set up (but do not evaluate) the integral expression for the area of the region enclosed by $y = e^x$, $y = \ln x$, and the lines $x = 1$ and $x = 2$.
- 8 Set up (but do not evaluate) the integral expression for the area between $y = x^3 - 4x$ and the x-axis on $[-2, 2]$, accounting for the curve crossing the axis at $x = -2, 0, 2$.
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Area between curves and a table of values

- 9 Functions f and g satisfy $f(x) \geq g(x)$ for all x in $[0, 6]$, with selected values given: $f(0) = 10$, $f(2) = 8$, $f(4) = 9$, $f(6) = 12$ and $g(0) = 3$, $g(2) = 4$, $g(4) = 5$, $g(6) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals to approximate the area between the curves, $\int_0^6 [f(x) - g(x)] dx$.